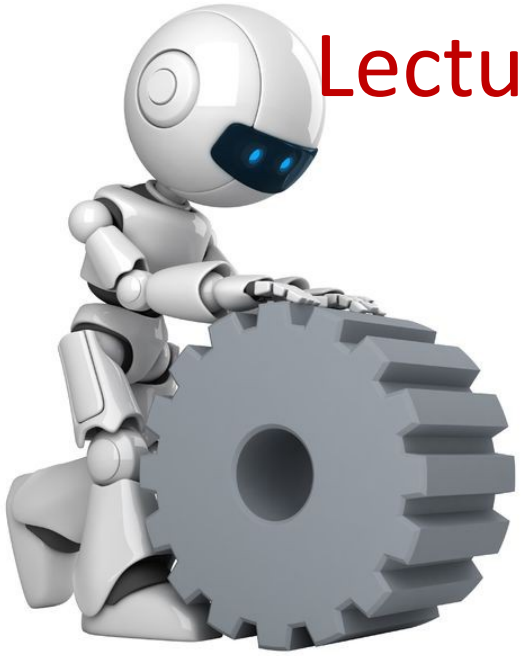
ICAT3110-3011

Intelligent Robotics



Lecture 12: Smart logistics and AI agents

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Guest session and project ideas

- In Lecture 12, we will have a special guest
- Thomas Hoglund, CEO of DONE Robotics Oy
- He will share very important insights about his robotics company
- Also, he will share project-work ideas that you can implement as Task (B)

Smart Logistics

- Logistics are **important drivers** for the competitiveness of industries and material supply.
- The development of smart logistics, powered by **precise positioning** and communication technologies can significantly improve the efficiency of logistics.
- Nowadays, smart logistics are considered a fundamental pillar of **Industry 4.0**.
- They enable companies to perform several **critical activities**, such as demand forecasting, sales planning and inventory management

Smart Logistics in Industry 4.0 Era

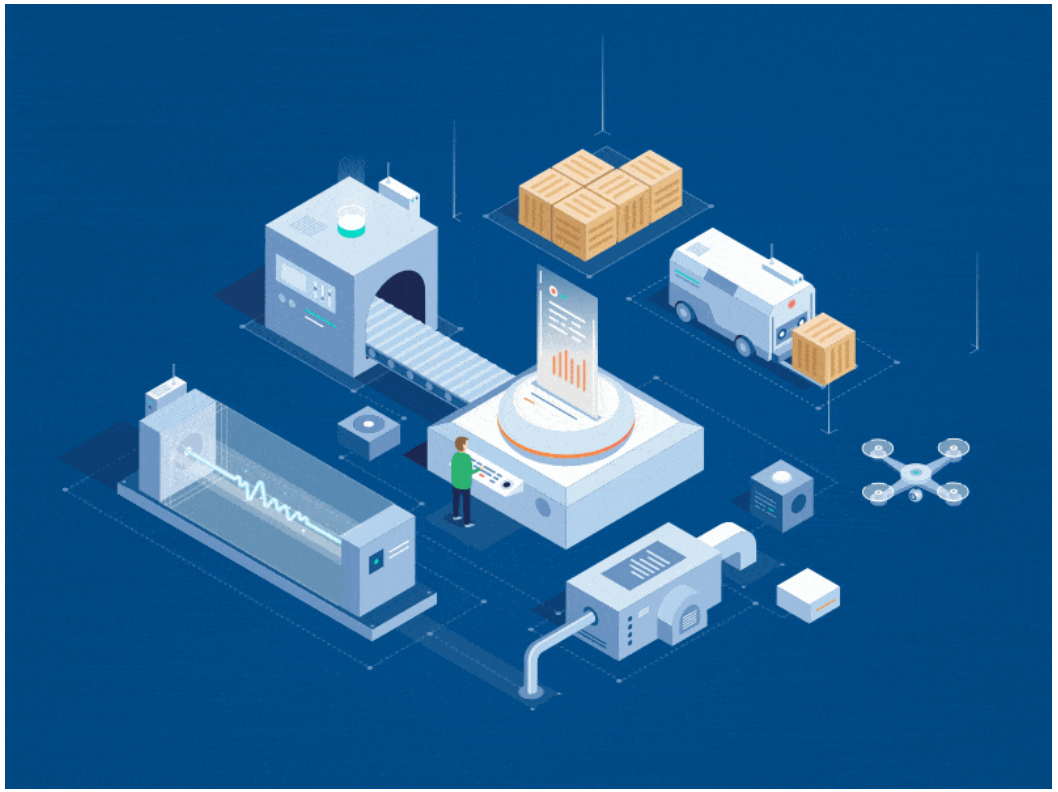

Image Source: Pinterest – @manoharparakh

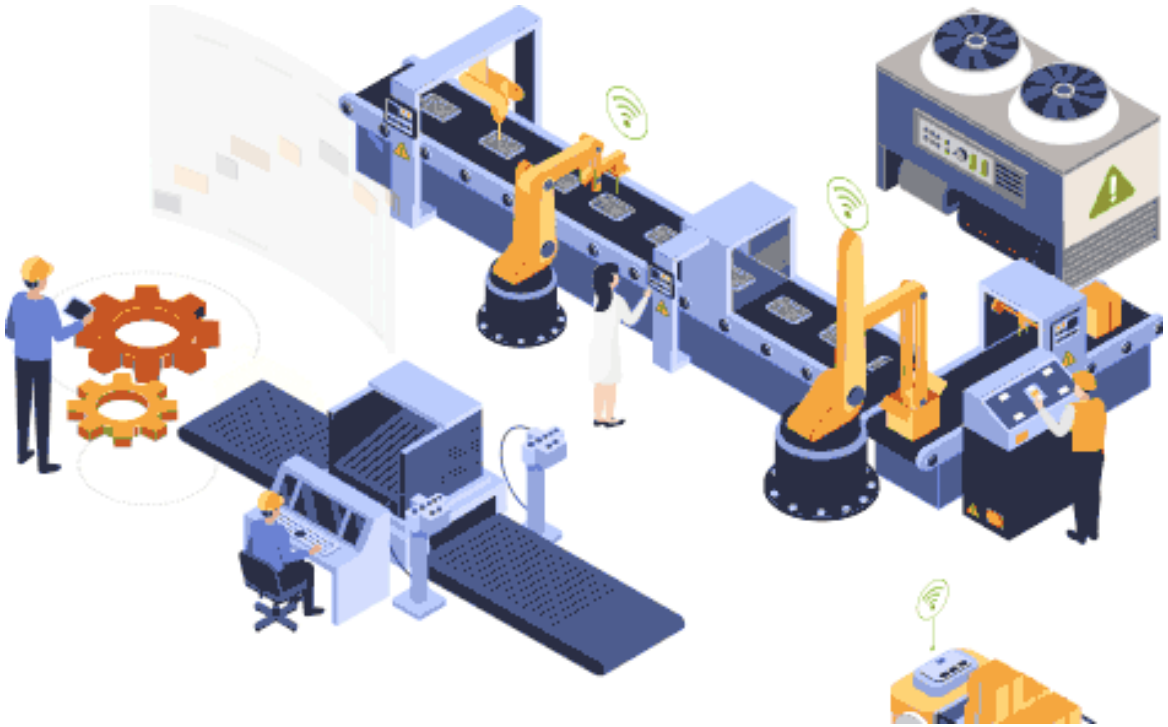

Image Source: LinkedIn – Patrick Mutabazi

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Indoors

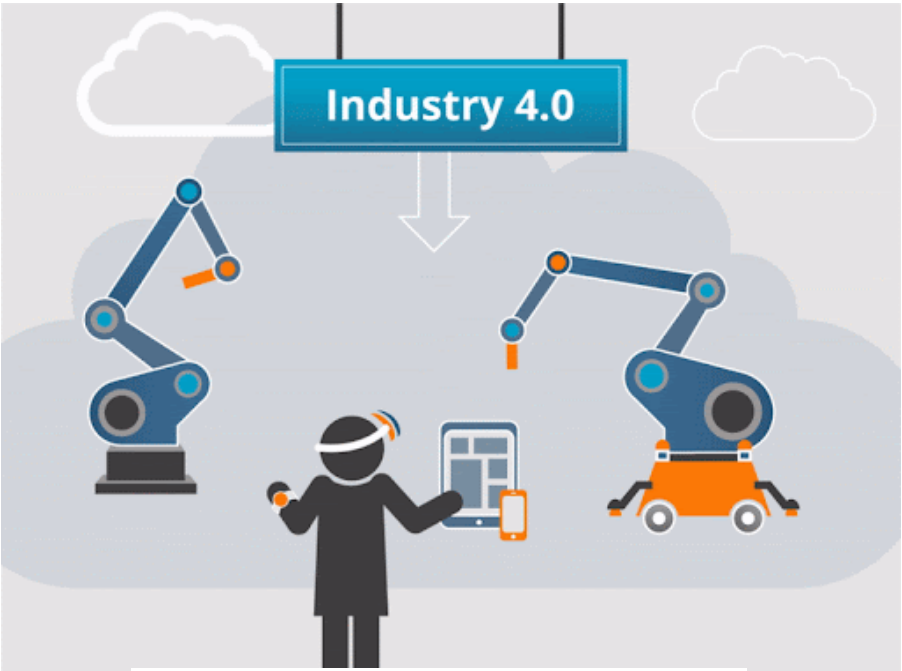


Image Source: LinkedIn – Patrick Mutabazi

Outdoors

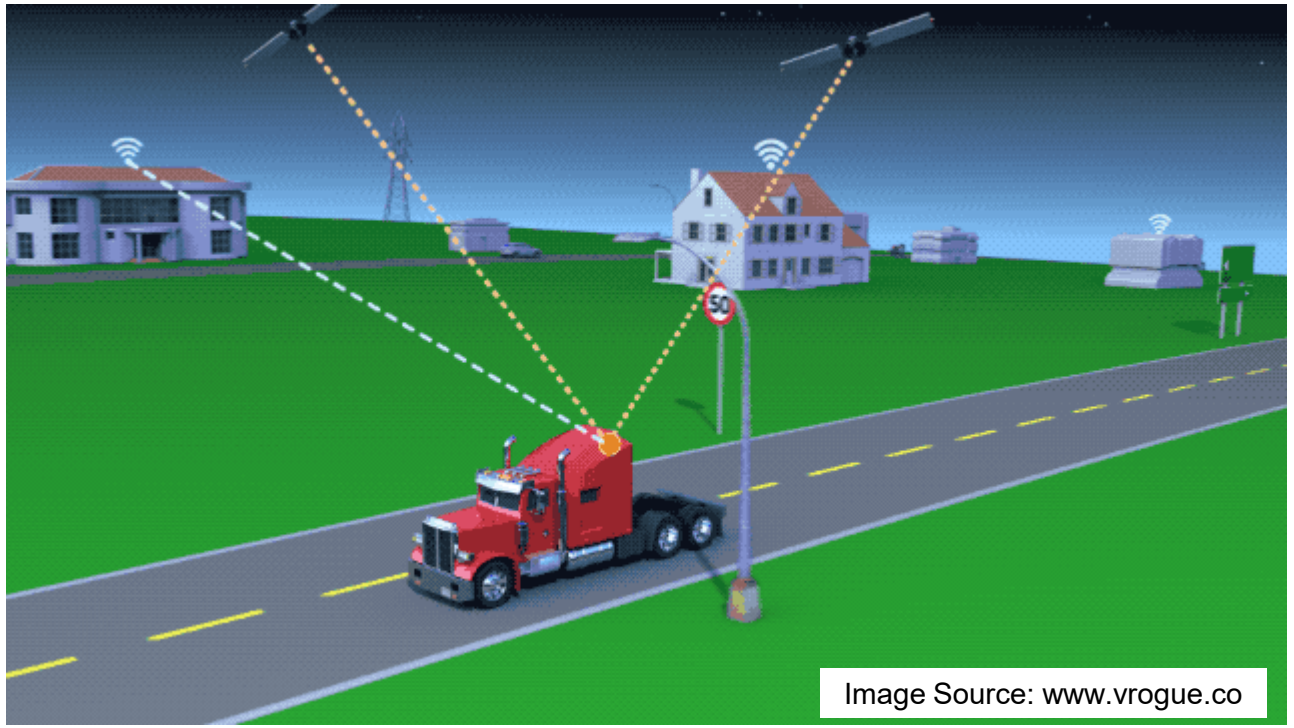
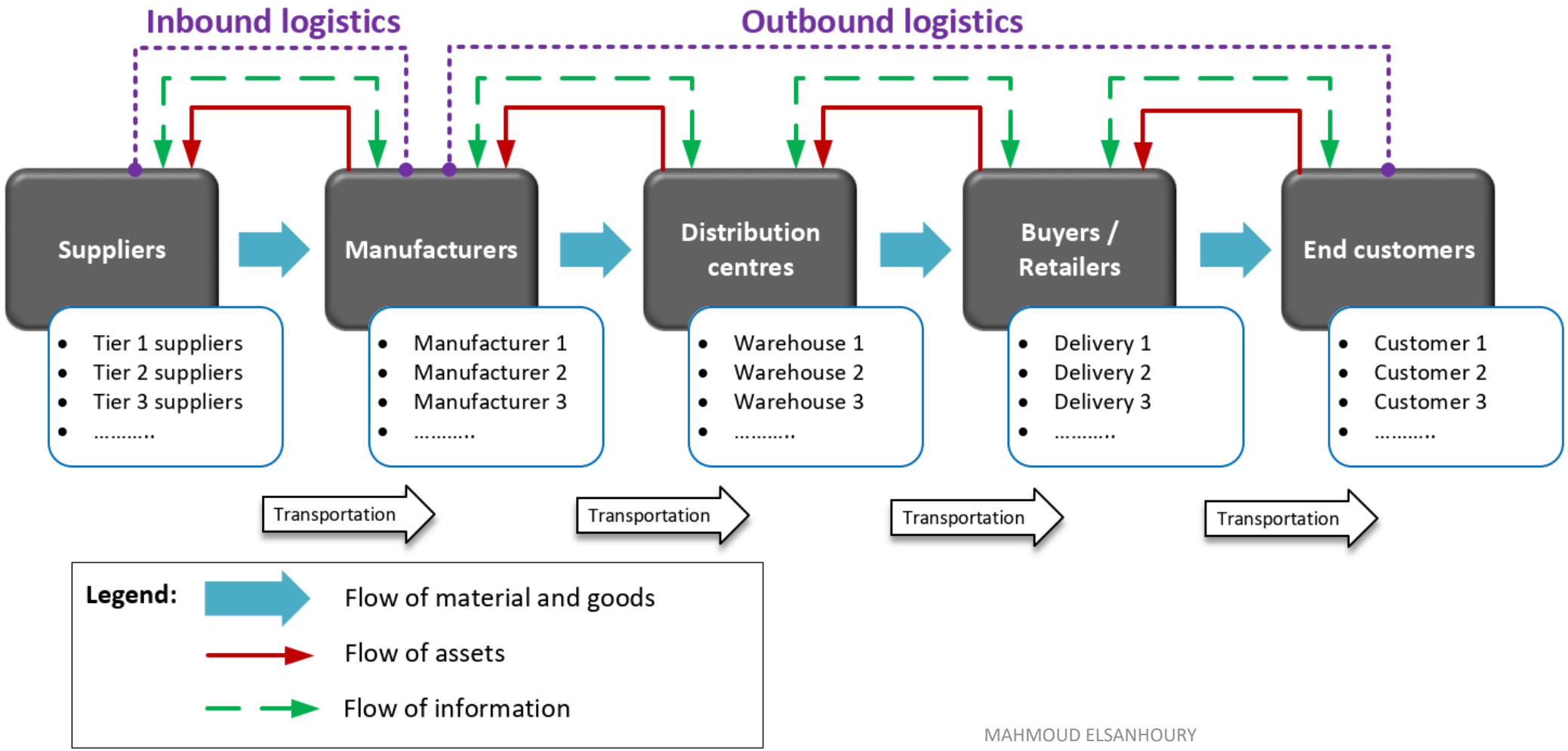
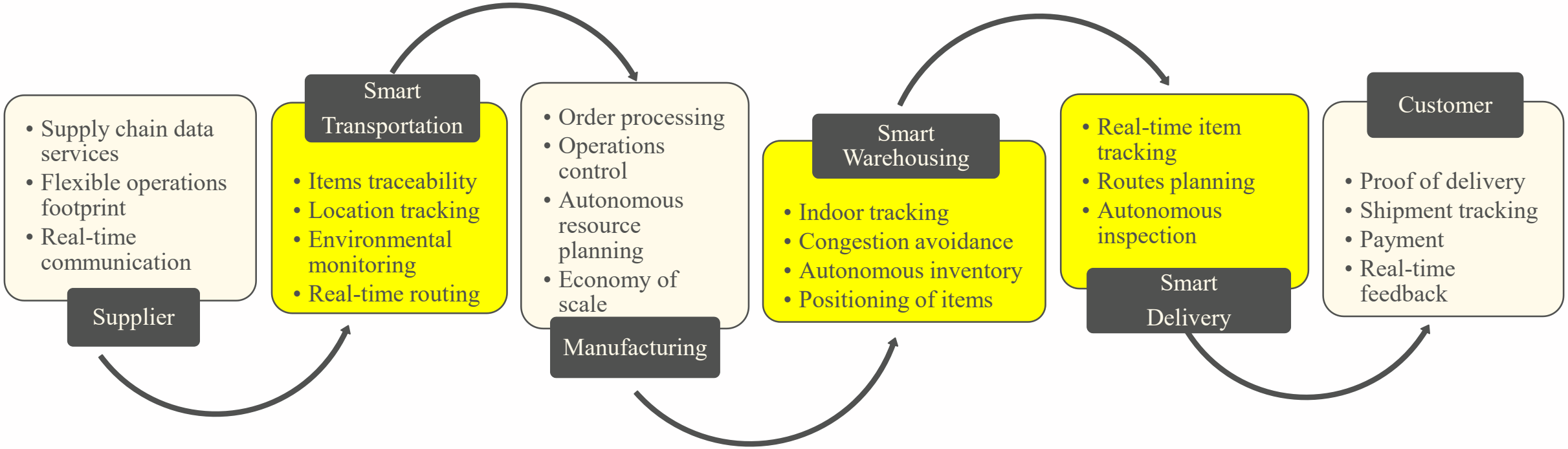


Image Source: www.vrogue.co

Smart logistics



Smart logistics



Fundamental elements and functionalities of smart supply and logistics chain.

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Bonus Point

We are living in the era of Industry 4.0, What were the key features of Industry 1.0, 2.0, and 3.0?

Bonus Point

We are living in the era of Industry 4.0, What were the key features of Industry 1.0, 2.0, and 3.0?

Answer

Industry 1.0: Mechanization using water and steam power.
Industry 2.0: Mass production using electricity and assembly lines.
Industry 3.0: Automation using electronics, computers, and IT.
Industry 4.0: Smart manufacturing using cyber-physical systems, IoT, AI, and data.

Smart Logistics - Definition

- Smart logistics are considered fairly **complex phenomena** that can be applied in geographically **dispersed areas** for tracking materials and goods with relatively low costs.
- Such a phenomenon is generally characterized by the introduction of new technologies such as: **IoT**, 5G, sensors, radio frequency identification (RFID) tags, smart products, actuators and **intelligent machines**.
- Smart logistics have the **ability**, **reliability**, **traceability** and **authenticity** of the information, which are useful to **establish intelligent** relationships among the stakeholders of the supply chains.

Smart Logistics - Stages

- Logistics can be divided into **six layers**:
  - (0) raw material (items),
  - (1) package,
  - (2) transport unit,
  - (3) unit load (pallet),
  - (4) container
  - (5) transportation unit (e.g. truck, ship and train).
- GNSSs are typically used for tracking large containers and transport equipment (the highest layers) in **outdoor** venues. On the other hand, the smaller cargo units (layers 1-3) are typically handled **indoors**, which can be tracked using indoor positioning technologies.

Smart Logistics - Technologies

- **Real-time** location tracking is increasingly attracting global logistics companies due to the need for visibility.
- To some extent, some technologies are deemed expensive and power-consuming. But, RFID is a **more available technology** to **track** the materials of **lower** level layers and items.

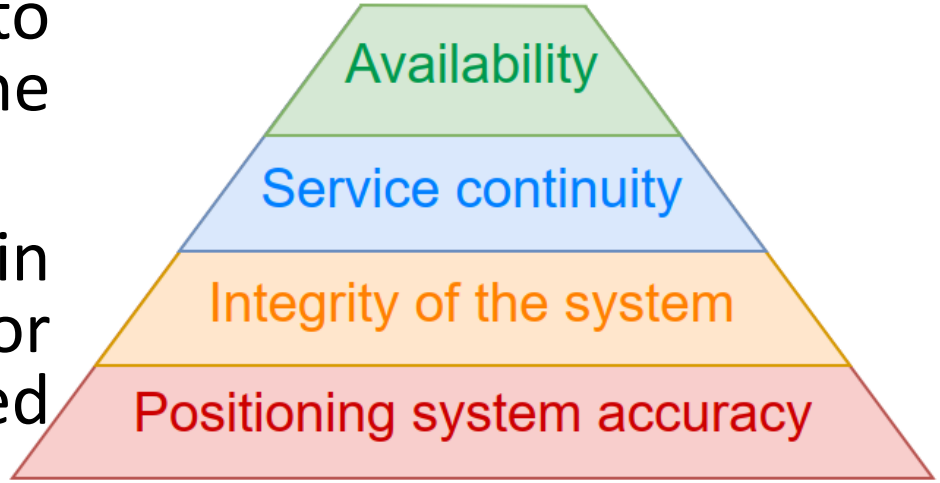
- However, the limited power source of **RFID** tags minimizes their operational range to a **couple of metres**. Therefore, they are mainly used for **identification** rather than positioning purposes.
- Similarly, **Bluetooth** and **Wi-Fi** are also used for indoor positioning, but their operating ranges are no more than 3-5 m.

Considerations for Smart Logistics

- In the case of autonomous robots operating in indoor environments (e.g. warehouses), **accurate positioning** is essential for indoor navigation. For outdoor smart logistics (e.g. smart delivery or shipping) that require reliable positioning and navigation systems, GNSSs are foreseen to be the best integrated positioning solutions for outdoor tasks.
- In the case of intelligent manufacturing, it is essential to **track** the **parts** along the production chain to take the right decisions. Real-time tracking of both stationary and moving parts in the production floor ensures safer operation with a reduced lead time.
- In addition to **parts**, it is also important to **track workers' movement** on the production floor to enhance operational flexibility.

Positioning Performance Metrics in Smart Logistics

- System **accuracy** is the degree of conformance of the estimated positioning values to the ground truth.
- The **integrity** of localization systems, is the trustworthiness of the information provided by the navigation engine.
- **Continuity** is the probability of the system to maintain the desired service level within the operation period.
- While **availability** is the percentage of time in which the navigation engine is up-running for positioning and can be used by its intended users.



Before we continue to the next topic …

AI Agents

THE NEW TREND IN THE INTELLIGENT WORLD

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Insights from a former course participant

- Sumit Sahoo, a MSc degree student
- Former participant in Intelligent Robotics Course **2025**

Sumit Sahoo

Student

- Currently working with a Robotics start-up in Oulu.
- His MSc thesis concerns about integrating LLMs and Robots

- He will talk about this area of research – short session

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Project,
Exercises,
And the Exam

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Continuing …

AI Agents

THE NEW TREND IN THE INTELLIGENT WORLD

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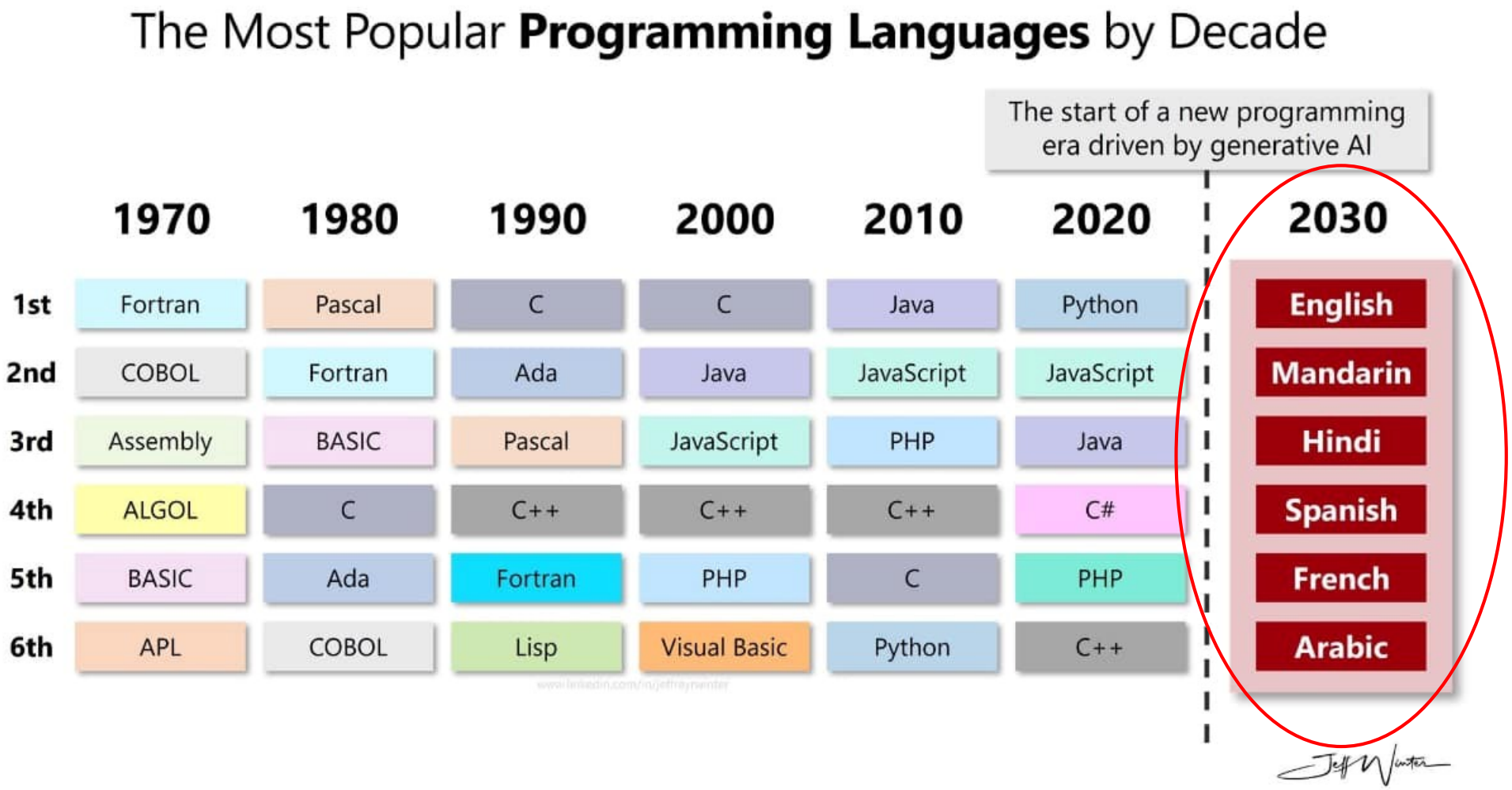
Emergence of Large Language Models (LLMs)

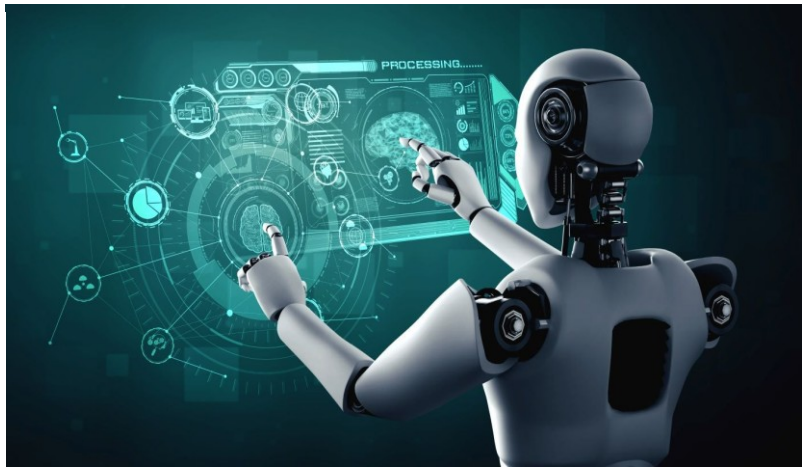
- **LLMs**, are trained on massive datasets to understand and generate human-like text.
- They excel in **language understanding**, text completion, and information synthesis.
- Models leverage **transformer** architectures with **billions** of **parameters**, enabling **complex pattern recognition**.
- This advancement **laid** the groundwork for autonomous systems that **interpret** and **interact** with the world **through <u>language</u>**.

Emergence of Large Language Models (LLMs)



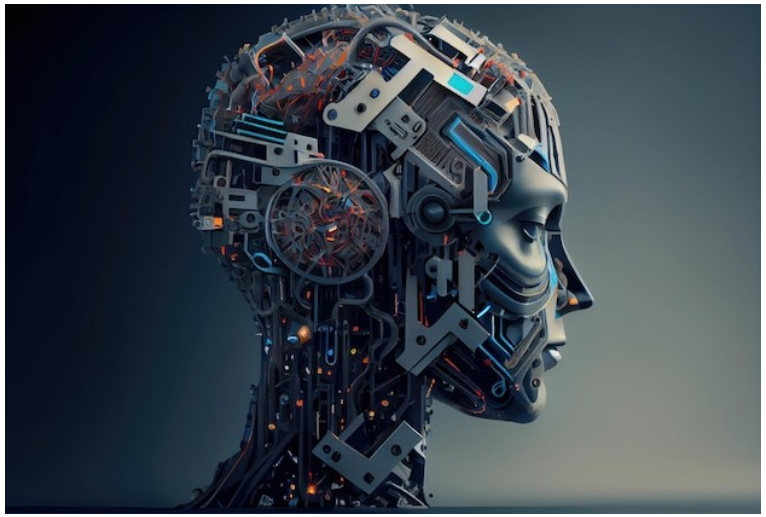
From Language Understanding to Autonomous Agents

- LLMs evolved **beyond** passive tools, now it is acting as **components** of larger, **decision-making** systems.
- New systems will use LLMs to plan, reason, and complete multi-step tasks.



- The **shift** from single-turn interactions to **continuous**, context-aware workflows **marks a step toward agency**.
- These **agents** can **explore** the internet, **learn** dynamically, and **refine** their strategies.

Cognitive Architectures and Agentic Design



- Agentic AI combines LLMs with **cognitive** models to simulate **human-like** thought processes.
- New agentic **architectures** can **inspire** modern designs, **enabling memory**, **reasoning**, and goal-directed behaviour.
- **Multi-agent** systems collaborate, **exchange** knowledge, and **self-improve** through **reinforcement** learning.
- This design paradigm **powers intelligent robotics** capable of handling **uncertain** and dynamic environments.

Role of Agentic AI in Intelligent Robotics

- **Robotics** systems **integrate agentic** AI to achieve **autonomous** navigation, manipulation, and interaction.
- Robots can **learn** from **real-world** feedback, **adapting** their **behaviour** to **new scenarios**.

- **Natural language** interfaces **allow** for **human-robot** collaboration, accelerating **problem-solving**.
- Examples include **autonomous vehicles**, service robots, and exploratory rovers.

Future of Agentic AI and Robotics

- **Future systems** may evolve into **lifelong** learners, continuously growing their knowledge and skills.
- **Agentic AI** could enable **collective intelligence**, where **distributed robots** share experiences globally.
- **Ethical considerations** and **safety** measures will guide development to prevent unintended consequences.
- Understanding and harnessing agentic AI will be **pivotal** for next-gen robotics research and real-world applications.

End of Lecture 12/12

Keep in touch ☺

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